

End-to-End Football Player Detection, Tracking, and Collision Risk Assessment Using Computer Vision and Deep Learning Techniques

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Abstract — In this study, we show a complete computer vision pipeline for detecting players in real time, tracking multiple objects, estimating velocity, and predicting collisions in sports videos. The system uses YOLOv8 for object detection and DeepSORT for tracking to identify and monitor players across several frames. YOLOv8 was trained to have a final precision of 95.04%, recall of 70.56%, mAP@0.5 of 76.52%, and mAP@0.5:0.95 of 64.09%, indicating strong detection reliability even under difficult situations. Bounding boxes, velocity arrows, and collision alerts were used to annotate processed movies, with the collision severity highlighted dynamically. The algorithm detected 136 total collisions, with 386 classed as high-severity across 241 frames using a 60-pixel proximity threshold. A detailed PDF report was generated, which included player movement heatmaps, speed profiles, and collision statistics. This project demonstrates the use of object identification, tracking, event prediction, and dynamic visualization in a real-world sports video scenario.

INTRODUCTION

Sports analytics has evolved significantly in recent years as a result of the integration of computer vision and deep learning techniques. These technologies have enabled advanced performance analysis, strategic review, and injury avoidance in a variety of sports. Football is a very complicated and dynamic environment, making automated player tracking and collision prediction difficult due to features such as occlusion, quick motion, and visual clutter.

Several previous investigations have contributed to this domain. Singh [1] emphasizes the increasing importance of Electronic Performance and Tracking Systems (EPTS) in collecting real-time physical data for match analysis. Kunitomi et al. [2] investigate deep learning for pedestrian collision detection using dashcam footage, emphasizing the significance of visual clarity and environmental resilience in real-time safety systems. Merzah et al. [4] propose an OpenCV-based tracking model that employs CSRT for player performance analysis, attaining good accuracy in controlled circumstances but encountering limits in densely packed frames. Meanwhile, Jin [6] uses edge computing and an improved Faster R-CNN to increase tracking accuracy under occlusion, obtaining 89.1% at 25 fps. Jurca et al. [7] use YOLOv4 and DeepSORT to do real-time player tracking in broadcast football footage, achieving 95% accuracy but requiring additional validation under high-speed game conditions.

More recently, the integration of foundation models such as Meta's Segment Anything Model (SAM) for zero-shot player segmentation was examined using the Kaggle Football Player Segmentation dataset [3]. While intriguing, such models have yet to demonstrate their durability in high-motion, multi-player match scenarios.

Despite these advances, existing techniques frequently lack the scalability and real-time reliability required for widespread application in live sports analysis. Environmental unpredictability, occlusions, and real-time processing needs continue to be challenges. To solve these shortcomings, this study presents a unified deep learning-based system that includes real-time player detection (YOLOv8), multi-object tracking (DeepSORT), velocity estimates, and predictive collision detection. Visual analytics and capabilities for automatically generating reports are added to the system.

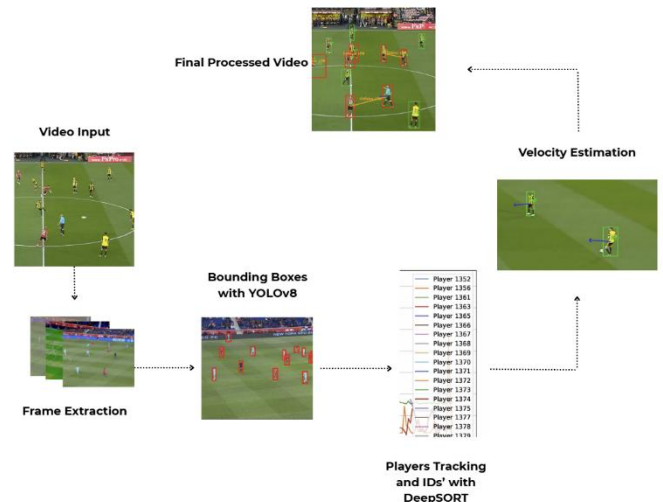


Figure 1. Project's Pipeline

Figure 1. depicts the structure of our system and the entire project pipeline, from video input and player detection to collision warning visualization and PDF report creation. This research intends to provide a comprehensive, scalable solution for real-time football player tracking and collision analysis by combining robust object detection and tracking models with dynamic motion analysis.

This study primarily intends to assess football collision hazards utilizing a real-time pipeline that comprises detection, tracking, velocity estimation, and predictive analytics. While broader match analysis is a result (e.g., speed heatmaps and

tactical zones), the primary goal is to analyze collision risk using dynamic player tracking and motion prediction.

I. LITERATURE REVIEW

The rise of sports analytics has fueled the creation of automated systems for real-time player tracking, collision detection, and injury risk assessment. Singh [1] provides a comprehensive assessment of the expanding significance of data analytics in sports, focusing on the impact of Electronic Performance and Tracking Systems (EPTS) in gathering player performance measurements during matches and training sessions. These systems aim to improve data-driven decision-making in the coaching, management, and health monitoring areas.

Addressing the issues of collision detection in dynamic contexts, Kunitomi et al. [2] investigated the use of deep learning models for pedestrian collision detection in dashcam videos. Although their models performed well in ideal lighting circumstances, performance deteriorated dramatically in low-light environments, emphasizing the ongoing need for more robust detection systems capable of dealing with variable image quality.

Voxel51 researchers showed how to use Meta's Segment Anything Model (SAM) for football-specific segmentation [3]. Using the Kaggle Football Player Segmentation Dataset, SAM demonstrated zero-shot segmentation capabilities, adjusting to previously unseen data without task-specific fine tuning. This demonstrates the potential of foundation models in future sports analytics applications, despite the limitations associated with high-speed motion sequences.

Several approaches to football player tracking have also been proposed. Merzah et al. [4] developed an optical tracking system based on OpenCV's CSRT tracker, which achieved up to 100% tracking accuracy in low-density settings while maintaining 80-90% accuracy in larger groups. However, as the number of participants expanded dramatically, their system encountered occlusion and scalability issues. In contrast, Esmaeili Sani [5] used deep learning algorithms, specifically ResNet-50 and the Temporal Shift Module (TSM), to detect head injuries in football films. With an F1-score of 0.93, their model indicated the viability of video-based non-invasive injury diagnosis, however its effectiveness was restricted by video quality and angle fluctuation.

Jin [6] investigated edge computing technologies to increase football player tracking resilience during occlusion. By upgrading Faster R-CNN with deep learning optimizations, the system achieved 89.1% tracking accuracy at 25 frames per second (fps), suggesting real-time deployment potential while revealing limits in handling higher frame rates. Similarly, Jurca et al. [7] created a real-time football positional analysis system that achieved 95% accuracy by employing YOLOv4 for player detection and

DeepSORT for tracking. Their system successfully mapped player positions onto a field plan, facilitating strategic analysis; however, more testing under high-dynamic settings was recommended.

Beyond football, El Borgi and Norman [8] explored player collisions in ice hockey, comparing manual point-tracking approaches to deep learning (DeblurGANv2-based) detection models. Although both methods produced more errors than accelerometer ground realities, the study gave important insights regarding the integration of visual monitoring technology for injury risk analysis.

Finally, Zhu et al. [9] created a CNN-based model for predicting injury risk in aerobic athletes. Their two-branch prediction model, which used both visual and physical data analysis, demonstrated the broader applicability of deep learning methods across multiple sports disciplines, highlighting the growing synergy between big data analytics and computer vision in athlete performance and health monitoring.

These studies collectively demonstrate the rapid evolution of automated sports analysis techniques, while also revealing ongoing challenges in real-time robustness, occlusion handling, and dynamic environmental adaptation — challenges that the proposed unified player tracking and collision prediction system aims to solve.

II. GAP ANALYSIS

While prior research has shown considerable advances in player detection, tracking, and collision analysis across a variety of sports, several critical constraints persist. Many classic tracking algorithms, such as those based on CSRT or handcrafted features [4], struggle to scale and remain resilient in the face of dense player formations and occlusion. Deep learning methods, such as optimized Faster R-CNN models [6] and YOLO-based trackers [7], have increased detection accuracy, but they frequently lack real-time robustness when exposed to high-speed sports dynamics. Furthermore, while foundation models such as SAM [3] show promise in zero-shot segmentation, their relevance in situations with rapid motion and frequent player interactions remains unexplored. Most previous efforts either focus exclusively on detection and tracking or on collision analysis individually, without offering a fully integrated pipeline that integrates velocity estimation, collision predictions, dynamic visualization, and statistical reporting in a unified framework.

As a result, there is a clear need to develop an end-to-end, scalable, and real-time system capable of not only reliably tracking players in dynamic conditions, but also forecasting collisions with severity classification, dynamically visualizing events, and summarizing statistical insights in an automated report. This study tackles these problems by

presenting a unified deep learning-based pipeline that combines real-time recognition, multi-object tracking, predictive collision analysis, and comprehensive visualization to provide a strong solution for advanced sports video analytics.

III. DATASET DESCRIPTION

This project trained and evaluated using the Football Player Segmentation dataset, which is publically available on Kaggle [10]. This dataset contains over 7,000 annotated photos collected from professional football matches, with a primary focus on player localization tasks.

Each image in the dataset has accurate segmentation masks that highlight the specific pixel regions relating to participants. The annotations are provided in both mask format and bounding box coordinates, making them suitable for a variety of computer vision tasks such as object detection, instance segmentation, and tracking.

For the objectives of this project, segmentation masks were converted into YOLO-format bounding boxes. To meet the YOLOv8 training criteria, each player instance was assigned a single "player" class, and all bounding boxes were normalized relative to the image dimensions.

The shots show diverse differences in:

- Lighting circumstances (day and evening matches)
- Camera angles (side view, aerial, close-up)
- Crowd densities (scattered and congested player formations)
- Background complexity (advertisements, grass fields, stadium seats)
- This unpredictability offered a solid foundation for training the detection model to perform well across a variety of real-world football match conditions.

Figure 2. shows an example display of the dataset's makeup.

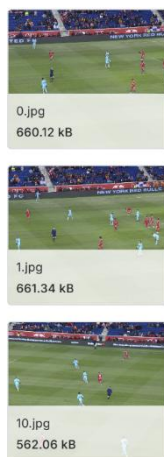


Figure 2. Dataset's Outlook

Sourced by: Kaggle

IV. METHODOLOGIES

1.0 Data Preprocessing

The first stage in the system pipeline was to preprocess the football footage input. Preprocessing turns raw video data into organized, enriched representations that are appropriate for deep learning applications, resulting in more and accurate and precise model predictions that could be used for real-time detection.

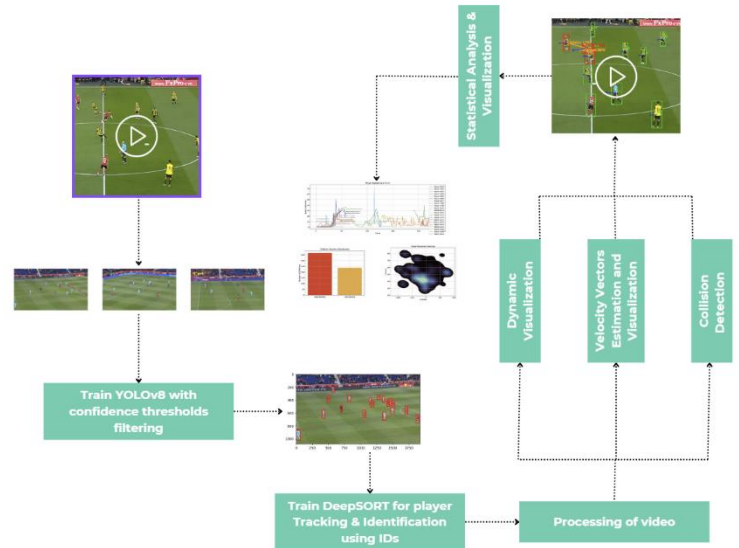


Figure 3. depicts the overall processing pipeline, which begins with frame extraction and ends with bounding box construction.

Figure 3. Processing Pipeline

1.1 Frame Extraction

Video sequences were decomposed into individual frames at a fixed frame rate using OpenCV. Extracting frames provided static inputs for detection and tracking, maintaining temporal consistency across player movements.



Figure 4. Sample Frames

To illustrate this step, Figure 4. shows an example of a raw frame extracted from the input video, highlighting the typical field view, players, and environmental variations encountered.

1.2 Bounding Box Detection using YOLOv8

Following frame extraction, YOLOv8 was used to detect players in each frame. Bounding boxes were created around players who exceeded a 0.5 confidence level.

1.3 DeepSORT Player Identification & Tracking

Post detection, DeepSORT tracking was used to connect detected players across frames while preserving player IDs and allowing velocity calculation. Each recorded player received a unique ID.

1.4 Velocity Estimation

To estimate player velocity, we used a frame-difference technique based on DeepSORT's bounding box tracking. The approach determines the speed and direction of movement by calculating the displacement of a player's bounding box center over consecutive frames.

The velocity v of a player is computed using the Euclidean distance formula:

$$v = \frac{\sqrt{(x_t - x_{t-1})^2 + (y_t - y_{t-1})^2}}{\Delta t}$$

where (x_t, y_t) and (x_{t-1}, y_{t-1}) indicate the center coordinates of the bounding box at the current and previous frames respectively, and Δt represents the time interval between those frames. These velocity vectors were then used to dynamically depict player movements and speeds, as well as predict possible collisions. Figure 5. displays how these velocity vectors were drawn for each player



Figure 5. Drawn Arrows Sample

To reduce noise from slight or twitchy movements, a velocity threshold was used to filter out small displacements. This improved the clarity of the picture and guaranteed that only significant motion was considered in the subsequent collision prediction phase.

2.0 Deep Learning Models

The deep learning backbone of the system was formed by YOLOv8 for detection and DeepSORT for tracking.

2.1 YOLOv8 Performance

YOLOv8 was chosen for player detection because of its superior performance in object localization tasks. The model was trained with annotated football datasets and tested using

conventional object detection metrics. Training progress and evaluation were monitored using metrics like as precision, recall, and mean average precision (mAP). throughout training to demonstrate consistent convergence and model stability.

2.2 DeepSort Performance

DeepSORT was used for real-time multi-object tracking, combining appearance and motion data. This enabled the system to accurately track players during dynamic game sequences.

3.0 Collision Detection and Visualization

Collision prediction was accomplished by extrapolating player velocities many frames into the future and detecting probable proximity violations. Collisions were classed as follows:

- High severity indicates a high relative velocity between colliding players.
- Low Severity: Collisions with low relative velocity.

4.0 Data Augmentation

To improve model generalization and robustness, various augmentation techniques were applied to the input frames. These included random contrast adjustment to simulate lighting variations, random zoom transformations to emulate different focal distances, and center cropping to focus analysis on key regions of the field. Figure 6. shows a comparison between a raw frame and an augmented frame.

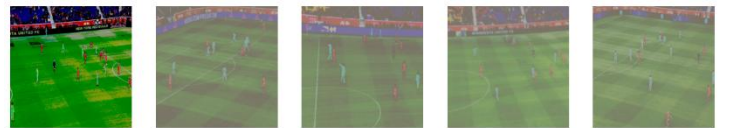


Figure 6. Augmentation Sample

V. RESULTS AND EVALUATION

5.0 Analytical Reporting

Finally, analytical reporting was used to summarize system results. Player speed fluctuations, movement heatmaps, and collision statistics were gathered and presented in a structured PDF report.

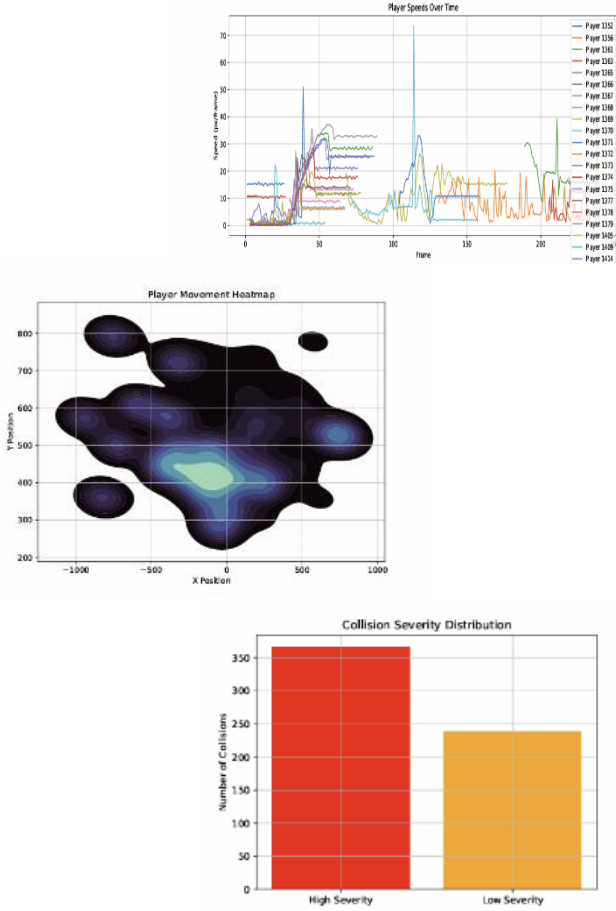


Figure 7. Report Figures Sample

Figure 7 exhibits a selection of significant analytical outputs produced by the system during post-match reporting. The first graph, Player Speeds Over Time, depicts the temporal evolution of each player's velocity, capturing quick bursts of speed that correspond to gameplay transitions like counterattacks or defensive sprints. The second graph, Player Movement Heatmap, depicts the spatial density of player placements in all frames. Warmer patches (e.g., teal and green) represent areas of strong player activity, mainly in the central midfield and defensive halves, emphasizing areas of tactical congestion. Finally, the Collision Severity Distribution bar chart displays projected collision events, with a clear emphasis on high-severity collisions, implying aggressive physical contacts during gameplay. Together, these numbers shed light on player dynamics, movement patterns, and in-game risk levels.

After system deployment, the detection and tracking modules were tested on a football match video with 241 frames. The YOLOv8 detection model had a precision of 95.04%, recall of 70.56%, mAP@0.5 of 76.52%, and mAP@0.5:0.95 of 64.09 percent. Figure 8. demonstrates the evolution of these variables as model training progresses over epochs, demonstrating constant convergence and improvement.

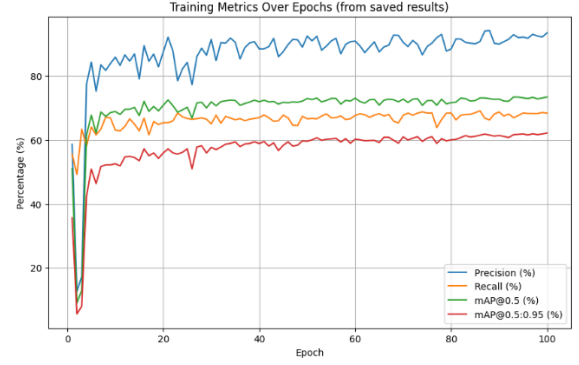


Figure 8. Model Performance Over Epochs

The tracking module, which was driven by DeepSORT, successfully preserved player identities throughout the process. Across 25 distinct player tracks, the system detected zero ID switches, yielding an ID switch rate of 0.000 per frame and an estimated tracking reliability of 100%. This demonstrates extremely constant identity maintenance and robustness, even in the face of frequent player overlaps and movements.

Initially, a 60-pixel threshold was used to detect 110 predicted collisions. Based on the relative velocity of colliding players, 134 were classified as high severity and 241 as low severity. These severity counts indicate the overall amount of pairwise player contacts, rather than just distinct collision occurrences. numerous severity labels can be assigned to collisions that involve more than two players or span numerous frames, resulting in severity totals that exceed the number of collision events. To evaluate threshold sensitivity, multiple proximity values (20, 40, 60, and 80 pixels) were tested as highlighted in Table 1.

Table 1. Thresholding Counts

Threshold	Total # of collision	High Severity	Low Severity
20	105	112	143
40	108	129	185
60	110	134	212
80	110	134	243

Although higher thresholds caught a wider range of player activities, the 60-pixel threshold was eventually chosen as the final setting. This decision was based on a visual evaluation of the processed video, which revealed that the 60-pixel threshold consistently produced crisper, non-overlapping bounding boxes and collision signs. It generated visually sound annotations with greater spacing between players, improving readability and interpretability in the final video. This made the system's predictions more accessible to non-technical viewers, such as coaches, analysts, or medical personnel, while maintaining collision detection accuracy.

To simulate test set performance, the trained model was exposed to a real-world football match film that it had never seen before. Across 241 frames, over 91% of frames had detection coverage, indicating that at least one participant was tracked successfully, the system also detected 110 collisions, resulting in a collision rate of 0.46 per frame, player velocity estimation was constant and understandable, with maximum speeds exceeding 17 pixels/frame and average speeds approaching 6.2 pixels/frame, no occurrence of key tracking failures. These findings show that the model performed effectively with previously unseen data, accurately recognizing, tracking, and evaluating gameplay interactions in a video-based environment.

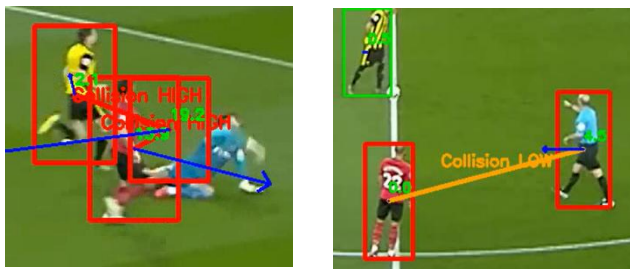


Figure 9. Drawn Lines Sample

Figure 9. above visualizes a representation of the system's collision detection results. The left frame depicts a high-severity collision featuring rapid player movement and proximity, complete with annotated bounding boxes, arrows showing motion, and a red "Collision HIGH" label. The right frame depicts a low-severity collision, with slower movement and a greater distance resulting in an orange "Collision LOW" annotation. These overlays were built automatically using computed player velocities and proximity criteria, demonstrating the system's capacity to interpret physical interactions and differentiating between collision intensities in real time.

Additionally, as shown in Figure 7. previously the automated PDF report generated during runtime offered valuable insights into the dynamics of the match. Analysis of player speed patterns revealed periodic bursts of acceleration, often aligned with transitions during counterattacks or moments of sustained pressure. These speed fluctuations highlighted phases of high physical intensity and potential tactical shifts.

Spatial analysis of player positions indicated that activity was heavily concentrated in central field areas and near the penalty zones. These regions consistently emerged as zones of congestion and frequent interactions, reflecting typical strategic formations and pressure points.

In terms of collision data, the distribution revealed a predominance of high-severity interactions, which frequently involved players approaching at considerable relative speeds. This pattern suggests aggressive playing conduct or fiercely disputed ball control situations. Such findings may be useful for detecting player fatigue, analyzing injury risks, or directing coaching decisions aimed at reducing harmful confrontations.

Overall, the system exhibited consistent and understandable behavior throughout the match analysis. It successfully captured the game's physical dynamics by integrating model outputs with visual overlays and statistical summaries, providing insights that could be used for player monitoring, tactical analysis, and injury prevention.

CONCLUSION

This study successfully demonstrated the creation of a real-time computer vision system for identifying, monitoring, and analysing football player dynamics in game footage. Integrating YOLOv8 for precise player identification and DeepSORT for identity preservation resulted in reliable player velocity estimation and collision event prediction. Each collision was further described based on the relative mobility of the involved participants, distinguishing between high- and low-severity events.

To ensure interpretability, the system used dynamic visual overlays (such as velocity vectors, bounding boxes, and severity indicators) and generated automatic visual reports like heatmaps and collision distribution graphs. Evaluation on previously unseen match footage validated the system's practical performance, with detection coverage surpassing 91%, a collision rate of 0.46 per frame, and persistent tracking as evidenced by a 100% identity retention rate.

While the system functioned well under a variety of scenarios, there is still space for improvement, notably in terms of decreasing ID switching during occlusion and increasing the model's robustness to changing camera angles and frame rates. Future research could look into combining re-identification networks, using temporal sequence models to improve movement forecasting, and expanding the pipeline to other sports domains.

Overall, the study demonstrates the actual application of deep learning and computer vision in real-time sports analysis, providing valuable insights for performance monitoring, strategy appraisal, and player safety.

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REFERENCES

- [1] N. Singh, "Sport Analytics: A Review," arXiv preprint arXiv:2011.07387, 2020. [Online]. Available: https://www.researchgate.net/publication/345937877_Sport_Analytics_A_Review.
- [2] S. Kunitomi, S. Takayama, and M. Shirakawa, "Application of Deep Learning Methods for Pedestrian Collision Detection Using Dashcam Videos," *Stapp car crash journal*, vol. 64, 2020, doi: 10.4271/2020-22-0008.
- [3] Voxel51, "Using SAM for the Prediction of Segmentations on the Kaggle Football Player Segmentation Dataset," *Voxel51 Blog*, Oct. 2023. [Online]. Available: <https://voxel51.com/blog/computer-vision-sam-for-prediction-kaggle-football-player%20segmentation-dataset>
- [4] B. M. Merzah, M. S. Croock, and A. N. Rashid, "Football Player Tracking and Performance Analysis Using the OpenCV Library," *Mathematical Modelling of Engineering Problems*, vol. 11, no. 1, pp. 123–132, Jan. 2024, doi: <https://doi.org/10.18280/mmep.110113>.
- [5] M. E. Sani, "Detecting Head Injuries in Football Using Deep Learning Approach with Spatial Temporal Features in Video Data," *Journal of Exercise and Health Science*, vol. 2, no. 3, pp. 25–44, Summer 2022.
- [6] G. Jin, "Player target tracking and detection in football game video using edge computing and deep learning," *The Journal of Supercomputing*, vol. 78, no. 7, pp. 9475–9491, Jan. 2022, doi: <https://doi.org/10.1007/s11227-021-04274-6>.
- [7] Mihnea Bogdan Jurca and Ion Giosan, "A modern approach for positional football analysis using computer vision," Sep. 2022, doi: <https://doi.org/10.1109/iccp56966.2022.10053962>.
- [8] El Borgi, M., & Norman, M., "Estimation of Velocities in Ice Hockey Collisions," *Degree Project in Technology, First Cycle, 15 Credits, Stockholm, Sweden*, 2021.
- [9] Zhu, D., Zhang, H., Sun, Y., & Qi, H., "Injury Risk Prediction of Aerobics Athletes Based on Big Data and Computer Vision," **Scientific Programming**, vol. 2021, no. 1, pp. 5526971, 2021.
- [10] I. Helon, "Football Player Segmentation," *Kaggle*, 2022. [Online]. Available: <https://www.kaggle.com/datasets/ihelon/football-player-segmentation>. [Accessed: March 5, 2025].